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## Pressure Drop in Pipe Fittings and Valves

### A Discussion of the Equivalent Length ( $L_e/D$ ), Resistance Coefficient (K) and Valve Flow Coefficient ( $C_v$ ) Methods

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#### 1. Introduction

The sizing of pipes for optimum economy requires that engineers be able to accurately calculate the flow rates and pressure drops in those pipes. The purpose of this document is to discuss the various methods available to support these calculations. The focus

will be on the methods for calculating the **minor losses** in pipe sizing and to consider in particular the following aspects:

- the advantages and disadvantages of each method
- Reynolds Number and the flow regime (turbulent vs laminar)
- the fitting size
- the roughness of the fitting
- the roughness of the attached piping
- converting data from one method to another

## 2. Background

Over the years excellent progress has been made in developing methods for determining the pressure drop when fluids flow through straight pipes. Accurate pipe sizing procedures are essential to achieve an economic optimum by balancing capital and running costs. Industry has converged on the Darcy-Weisbach method, which is remarkably simple considering the scope of applications that it covers.

The Darcy-Weisbach formula is usually used in the following form:

$$h_L = \left( \frac{fL}{D} \right) \left( \frac{v^2}{2g} \right) \quad \text{Equation (1)}$$

Equation (1) expresses the pressure loss due to friction in the pipe as a head ( $h_L$ ) of the flowing fluid.

The terms and dimensions in Equation (1) are:

|       |                                                                                           |
|-------|-------------------------------------------------------------------------------------------|
| $h_L$ | head of fluid, dimension is length                                                        |
| $f$   | Moody friction factor (also called Darcy-Weisbach friction factor), dimensionless         |
| $L$   | length of straight pipe, dimension is length                                              |
| $D$   | inside diameter of pipe, dimension is length                                              |
| $v$   | average fluid velocity (volumetric flow / cross sectional area), dimension is length/time |
| $g$   | acceleration due to earth's gravity, dimension is length/time <sup>2</sup>                |

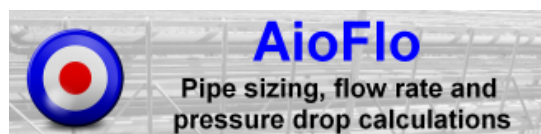
The dimensions in Equation (1) can be in any consistent set of units. If the Fanning friction factor is used instead of the Moody friction factor then  $f$  must be replaced by  $4f$ .

In long pipelines most of the pressure drop is due to the friction in the straight pipe, and the pressure drop caused by the fittings and valves is termed the "minor loss". As pipes get shorter and more complicated the proportion of the losses due to the fittings and valves gets larger, but by convention are still called the "minor losses".

Over the last few decades there have been considerable advances in the accurate determination of the minor losses, but as of now they cannot be determined with the same degree of accuracy as the major losses caused by friction in the straight pipe. This situation is aggravated by the fact that these recent developments have not filtered through to all levels of engineering yet, and there are many old documents and texts still around that use older and less accurate methods. There is still considerable confusion amongst engineers over which are the best methods to use and even how to use them.

Unfortunately one of the most widely used and respected texts, which played a major role in advancing the state of the art, has added to this confusion by including errors and badly worded descriptions. (See section 4 below)

Nevertheless, by employing the currently available knowledge and exercising care the minor losses can be determined with more than sufficient accuracy in all but the most critical situations.



### 3. The Three Methods for Minor Loss Determination

The 3 methods which are used to calculate the minor losses in pipe sizing exercises are the equivalent length ( $L_e/D$ ), the resistance coefficient ( $K$ ) and the valve flow coefficient ( $C_v$ ), although the  $C_v$  method is almost exclusively used for valves. To further complicate matters, the resistance coefficient ( $K$ ) method has several levels of refinement and when using this procedure it is important to understand how the  $K$  value was determined and its range of applicability. There are also several definitions for  $C_v$ , and these are discussed below.

For all pipe fittings it is found that the losses are close to being proportional to the second term in Equation (1). This term ( $v^2/2g$ ) is known as the "velocity head". Both the equivalent length ( $L_e/D$ ) and the resistance coefficient ( $K$ ) method are therefore aimed at finding the correct multiplier for the velocity head term.

#### 3.1 The equivalent length method ( $L_e/D$ )

This method is based on the observation that the major losses are also proportional to the velocity head. The  $L_e/D$  method simply increases the multiplying factor in Equation (1) (i.e.  $fL/D$ ) by a length of straight pipe (i.e.  $L_e$ ) which would give rise to a pressure drop equivalent to the losses in the fittings, hence the name "equivalent length". The multiplying factor therefore becomes  $f(L+L_e)/D$ .

In the early stages of a design when the exact routing of the pipeline has not been decided, the equivalent length can be estimated as a broad brush allowance like "add 15% to the straight length to cover the fittings". However, if the design is complete and a detailed take-off of the fittings is available a more accurate calculation of the minor losses is possible by using experimentally determined equivalent lengths for each of the fittings and valves.

It has been found experimentally that if the equivalent lengths for a range of sizes of a given type of fitting (for example, a 90° 1/2 long radius bend) are divided by the diameters of the fittings then an almost constant ratio (i.e.  $L_e/D$ ) is obtained. This makes the tabulation of equivalent length data very easy, because a single data value is sufficient to cover all sizes of that fitting. Some typical data is shown in the table below for a few frequently used fittings:

| Fitting Type             | $L_e/D$ |
|--------------------------|---------|
| Gate valve, full open    | 8       |
| Ball valve, full bore    | 3       |
| Ball valve, reduced bore | 25      |
| Globe valve, full open   | 320     |
| 90° 1/2 screwed elbow    | 30      |
| 90° 1/2 long radius bend | 13      |
| 45° 1/2 screwed elbow    | 16      |
| 45° 1/2 long radius bend | 10      |
| Welded Tee, thru-run     | 10      |
| Welded Tee, thru-branch  | 60      |

**Table of Equivalent Lengths for Pipe Fittings  
(Clean commercial steel pipe)**

This data is for illustration only and is not intended to be complete. [Comprehensive tables of Equivalent Length Values](#) for steel and plastic pipe are available in another of our articles.

Note that this fortuitous situation of having a constant  $L_e/D$  for all sizes does not apply to some fittings such as entrances and exits, and to fittings such as changes in diameter and orifices - both of which involve more than one bore size.

The equivalent length method can be incorporated into the Darcy-Weisbach equation and expressed in mathematical form as:

$$h_L = f \left( \frac{L}{D} + \sum \frac{L_e}{D} \right) \left( \frac{v^2}{2g} \right) \quad \text{Equation (2)}$$

Note that the expression  $\Sigma(L_e/D)$  is also multiplied by the Moody friction factor  $f$ , because it is being treated just as though it were an additional length of the same pipe.

The pipe length,  $L$ , in Equation (2) is the length of the straight pipe only. Some authors recommend that  $L$  include the flow distance through the fittings but this is wrong. The  $(L_e/D)$  factor is based on the overall pressure drop through the fitting and therefore includes any pressure drop due to the length of the flow path. The error is small and usually well within the tolerance of the data, so trying to measure all the flow path lengths is just a waste of time, as well as being technically wrong.

The applicability of the equivalent length  $(L_e/D)$  data to the laminar flow regime will be considered in section 3.4.3 below.

### 3.2 The resistance coefficient (K) method (sometimes called the "loss coefficient" method)

This method can be incorporated into the Darcy-Weisbach equation in a very similar way to what was done above for the equivalent length method. In this case a dimensionless number ( $K$ ) is used to characterise the fitting without linking it to the properties of the pipe. This gives rise to:

$$h_L = \left( \frac{fL}{D} + \sum K \right) \left( \frac{v^2}{2g} \right) \quad \text{Equation (3)}$$

Note that in this case the sum of the resistance coefficients ( $\Sigma K$ ) is not multiplied by the Moody friction factor  $f$ . Early collections of resistance coefficient ( $K$ ) values (for example the 3<sup>rd</sup> Edition of Perry's Chemical Engineers' Handbook in 1950) gave single values for each type of fitting, with the intention that the value be applicable to all sizes of that fitting. As more research was done it was found that in general the resistance coefficient ( $K$ ) decreased as the fitting size increased, and when the Hydraulic Institute published the "Pipe Friction Manual" in 1954 the coefficients were given in the form of graphs covering a wide range of sizes.

Up until that point in time the derived  $K$  values were for use in the fully turbulent flow regime only, and the 3<sup>rd</sup> Edition of Perry's Handbook makes specific mention of the non-applicability of the data to laminar (or viscous) flow.

The valve manufacturer, Crane Company, had been producing technical information for flow calculations since 1935 and launched their Technical Paper No. 410 "Flow of Fluids through Valves, Fittings and Pipe" in 1942. Since then this document has been regularly updated and is probably the most widely used source of piping design data in the English speaking world. The 1976 edition of Crane TP 410 saw the watershed change from advocating the equivalent length  $(L_e/D)$  method to their own version of the resistance coefficient ( $K$ ) method. This is widely referred to in the literature as the "Crane 2 friction factor" method or simply the "Crane  $K$ " method. Crane provided data for an extensive range of fittings, and provided a method for adjusting the  $K$  value for the fitting size. Unfortunately this welcome advance introduced a significant error and much confusion. The details of the Crane method, plus the error and source of the confusion are discussed separately in section 4 below.

By the time the 4<sup>th</sup> Edition of Perry's Handbook was published in 1963 some meagre data was available for resistance coefficients in the laminar flow regime, and they indicated that the value of  $K$  increased rapidly as the Reynolds Number decreased below 2000. The first comprehensive review and codification of resistance coefficients for laminar flow that I am aware of was done by William Hooper (1981). In this classic paper Hooper described his two- $K$  method which included the influence of both the fitting size and the Reynolds Number, using the following relationship:

$$K = \frac{K_1}{Re} + K_\infty \left( 1 + \frac{1}{D} \right) \quad \text{Equation (4)}$$

In this Equation  $K_\infty$  is the "classic"  $K$  for a large fitting in the fully turbulent flow regime and  $K_1$  is the resistance coefficient at a Reynolds Number of 1. Note that although the  $K$ 's and  $Re$  are dimensionless the fitting inside diameter ( $D$ ) must be given in inches.

The advances made by Hooper were taken a step further by Ron Darby in 1999 when he introduced his three- $K$  method. This is the method used in the [AioFlo pipe sizing calculator](#). The three- $K$  equation is slightly more complicated than Hooper's two- $K$  but is able to fit the available data slightly better. This equation is:

$$K = \frac{K_m}{Re} + K_i \left( 1 + \frac{K_d}{D^{0.3}} \right) \quad \text{Equation (5)}$$

In Equation (5) the fitting diameter (D) is again dimensional, and must be in inches. Possibly because of the significant increase in computational complexity over the equivalent length ( $L_e/D$ ) and Crane K methods, the two-K and three-K methods have been slow to achieve much penetration in the piping design world, apart from their use in some high-end software where the complexity is hidden from the user. Also, both of these methods suffered from typographic errors in their original publications and some effort is required to get reliable data to enable their use, adding to the hesitation for pipe designers to adopt them.

This slow take-up of the new methods is reflected in the fact that Hooper's work from 1981 did not make it into the 7<sup>th</sup> Edition of Perry's Handbook in 1997 (which still listed "classic" K values with no correction for size or flow regime). However, it is only a matter of time until some multi-K form becomes part of the standard methodology for pipe sizing.

The performance of the two-K and three-K methods can be compared over a range of pipe sizes by considering water flowing through a standard radius 90 degree elbow at a rate to give a pressure drop in straight pipe of the same diameter of 3 psi per 100 ft. For this exercise the coefficients for the two formulas were taken as

Hooper two-K:  $K_1 = 800$ ,  $K_\infty = 0.25$

Darby three-K:  $K_m = 800$ ,  $K_i = 0.091$ ,  $K_d = 4.0$

| Pipe Size<br>inch | 2-K<br>K-Value | 3-K<br>K-Value | Diff %<br>(2K-3K) |
|-------------------|----------------|----------------|-------------------|
| 1/4               | 1.096          | 0.743          | 38.4              |
| 1/2               | 0.715          | 0.574          | 21.9              |
| 3/4               | 0.593          | 0.516          | 13.8              |
| 1                 | 0.501          | 0.463          | 8.0               |
| 2                 | 0.379          | 0.392          | -3.3              |
| 3                 | 0.336          | 0.355          | -5.7              |
| 4                 | 0.315          | 0.333          | -5.7              |
| 6                 | 0.293          | 0.304          | -3.9              |
| 8                 | 0.282          | 0.287          | -1.7              |
| 10                | 0.276          | 0.274          | 0.6               |
| 12                | 0.271          | 0.264          | 2.6               |
| 14                | 0.269          | 0.260          | 3.7               |
| 16                | 0.267          | 0.253          | 5.4               |
| 18                | 0.265          | 0.247          | 7.0               |
| 20                | 0.264          | 0.242          | 8.4               |
| 24                | 0.261          | 0.234          | 11.0              |
| 30                | 0.259          | 0.224          | 14.5              |
| 36                | 0.257          | 0.217          | 17.0              |

**Table Comparing K-Values for Hooper 2-K and Darby 3-K Methods  
(Values are for std radius 90 deg bend in turbulent flow)**

This table shows that for piping sizes between 1" and 24" as typically used in process plants the differences between these two methods are small. What little experimental data has been published shows larger variations than the differences between these two methods, and suggests that both these methods are slightly conservative.

### 3.3 The valve flow coefficient ( $C_v$ ).

As the name suggests, this method is predominantly used in calculations for valves, but as will be seen later in this article it is easy to convert between  $C_v$  and resistance coefficient (K) values so it is possible to define a  $C_v$  for any fitting.

By definition, a valve has a  $C_v$  of 1 when a pressure of 1 psi causes a flow of 1 US gallon per minute of water at 60°F (i.e. SG = 1) through the valve. Since the pressure drop through a valve is proportional to the square of the flow rate the relationship between  $C_v$ , flow rate and pressure drop can be expressed as:

$$Q = C_v \sqrt{\Delta P / SG} \quad \text{Equation (6)}$$

This is a dimensional formula and the dimensions must be in the following units

|            |                                                      |
|------------|------------------------------------------------------|
| $Q$        | volumetric flow rate in US gallon per minute         |
| $\Delta P$ | pressure drop in psi                                 |
| $SG$       | specific gravity of liquid relative to water at 60°F |

In Britain a similar expression is used to define a  $C_v$  which is given in terms of **Imperial** gallons per minute, but using the same units for pressure drop and SG as in the USA. Great care has to be taken when using  $C_v$  values from valve manufacturers' catalogs to ascertain which basis was used in the definition.

In continental Europe valves were traditionally rated with a valve coefficient designated as  $K_v$ . This is also a dimensional formula and the units are as defined below:

$$Q' = K_v \sqrt{\Delta P' / SG'} \quad \text{Equation (7)}$$

|             |                                                      |
|-------------|------------------------------------------------------|
| $Q'$        | volumetric flow rate in cubic metres per hour        |
| $\Delta P'$ | pressure drop in kgf/cm <sup>2</sup>                 |
| $SG'$       | specific gravity of liquid relative to water at 15°C |

However, an updated definition is also used in Europe which has finally brought the valve coefficient into the modern era with SI Units. At present this definition is not widely used, but as more and more contractual documents encourage the use of SI Units it can be expected to grow in popularity. This coefficient is called the "Area Coefficient" and is written as  $A_v$ . Its definition is:

$$Q'' = A_v \sqrt{\Delta P'' / \rho} \quad \text{Equation (8)}$$

|              |                                                       |
|--------------|-------------------------------------------------------|
| $Q''$        | volumetric flow rate in cubic metres per second       |
| $\Delta P''$ | pressure drop in pascal ( $\equiv$ N/m <sup>2</sup> ) |
| $\rho$       | density of liquid in kg/m <sup>3</sup>                |

### 3.4 Comparison of the equivalent length ( $L_e/D$ ) and the resistance coefficient ( $K$ ) methods

As mentioned earlier, both these methods use a multiplier with the velocity head term to predict the pressure drop through the fitting. There is therefore no real difference between the two and provided that accurate characterizing data for the fitting is used, both methods can give equally accurate results.

By comparing Equations (2) and (3) we can see that the constants for the two methods are directly related by:

$$K = \frac{f L_e}{D} \quad \text{Equation (9)}$$

$$\frac{K}{f} = \frac{L_e}{D}$$

Thus, in any specific instance where all the fluid and piping details are known it is possible to get an exact conversion between the constants for the two methods. However, when engineers talk of comparing these two methods the real questions are related to how a  $K$  value or an  $L_e/D$  value obtained under one set of circumstances can be employed under a different set of circumstances.

These changed circumstances relate mainly to pipe material, fitting size, flow regime (ie Reynolds Number) and the roughness of the fitting itself.

#### 3.4.1 Effect of pipe material

The roughness of the piping attached to the fitting has no influence on the pressure drop through the fitting. However, because the equivalent length ( $L_e/D$ ) method expresses the pressure drop through the fitting in terms of the pressure drop through the attached piping, the pipe roughness does affect the length of piping that would have a pressure drop equivalent to the fitting. This is best illustrated with an example:

A flow rate of 150 USgpm through a 3" globe valve with a  $C_v$  of 105 (US units) would result in a pressure drop of 2.05 psi (using Equation (6)). This pressure drop would not be affected by the roughness of the pipe attached to it. If the piping were galvanized steel with a roughness of 0.006" the pressure drop in the pipe would be 2.72 psi per 100 ft. The length of galvanized piping that would give an equivalent pressure drop to the valve would be 75 ft, giving an  $L_e/D$  ratio of 290. If the piping were smooth HDPE with a roughness of 0.0002" the pressure drop in the pipe would be only 1.89 psi per 100 ft and the length of HDPE piping that would give an equivalent pressure drop to the valve would be 108 ft, giving an  $L_e/D$  ratio of 420.

In order to be able to use the equivalent length method as given in Equation (2) the  $L_e/D$  values used should strictly be relevant to the roughness of the piping in use. In practice the differences are often not important because of the "minor" nature of the pressure drop through the fittings. In the example given here the difference is 44%, and if this applies to the minor loss which is (say) 15% of the overall loss the effective error in the pipeline pressure drop is only 7% and this could well be within the overall tolerance of the calculation.

Nevertheless, it is best to be aware of how reported  $L_e/D$  values were obtained and to what piping they can be applied. Unfortunately the  $L_e/D$  values listed in texts do not usually mention the piping material, but in most cases it will be clean commercial steel pipe. The inability of the equivalent length method to automatically cope with changes in pipe roughness is a disadvantage of this method. We have prepared [comprehensive tables of Equivalent Length Values](#) as a function of pipe roughness in another of our articles.

The resistance coefficient (K) method is totally independent of the pipe roughness and the material of the attached piping is irrelevant when this method is used to calculate minor losses.

#### 3.4.2 Effect of fitting size

In section 3.1 it was noted that it has been found that the  $L_e/D$  ratio remains almost constant for a range of sizes of a given type of fitting. On the other hand, it was noted in section 3.2 that in general the resistance coefficient (K) values decreases with increasing fitting size. For the relationship of  $K/f = L_e/D$  from Equation (9) to apply it must mean that  $K/f$  remains constant, or that  $K$  and  $f$  change at the same rate. This observation was the basis of the Crane K method and is discussed further in section 4 below.

When using the equivalent length method, the ( $L_e/D$ ) ratio is multiplied by the friction factor and since the friction factor decreases as the pipe size increases the term ( $fL_e/D$ ) decreases accordingly. This makes the equivalent length method largely self-correcting for changes in fitting size and makes it very suitable for preliminary or hand calculations where ultimate accuracy is not the main goal.

The best available method available at present to accommodate changing pipe sizes appears to be Darby's 3-K method. This method predicts resistance coefficients slightly higher than some of the older data that did take fitting size into account (for example, the Hydraulic Institute "Pipe Friction Manual") but because it is given in algebraic form it is much easier to use in modern spreadsheets and computer programs than the graphical data presented in the older documents.

As an illustration, consider 2" and 20" long radius bends in a clean commercial steel pipeline. At fully turbulent flow the resistance coefficient (K) calculated by the Darby method would be 0.274 for the 2" bend and 0.173 for the 20". This is a 37% decrease. If the equivalent length is calculated from these K values and from the Moody friction factor for clean commercial steel pipe then the 2" bend has an ( $L_e/D$ ) value of 13.8 and the 20" bend has value of 14.0 - a change of just over 1% and a strong recommendation for the equivalent length method.

#### 3.4.3 Effect of flow regime (Reynolds Number)

The early "classic" K values were measured under fully turbulent flow conditions. This is the flow regime most often used in industrial applications and it was an understandable place to start accumulating data. But it was observed that at lower Reynolds Numbers in the transition zone between  $Re = 4,000$  and fully developed turbulent flow the K values did increase somewhat. When the investigations were extended into the laminar regime very large K value increases were found.

Continuing with the example of the long radius bends, at a Reynolds number of 100 the Darby 3-K method predicts that both the 2" and the 20" L.R. bends would have K values of 8.2. This is a huge increase over the turbulent flow situation. It should be remembered though that in the laminar flow regime velocities tend to be very low, making the velocity head ( $v^2/2g$ ) low and since the pressure drop is calculated as the product of the K value and the velocity head, the effect of the increase in K is partially offset and the pressure drop can be low in absolute terms.

Again, the equivalent lengths can be calculated from these K values and the Moody friction factors to give an  $(L_e/D)$  ratio and this turns out to be 12.8 for both bends. This small change in the  $(L_e/D)$  ratio compared with those found in section 3.4.2, despite such a large change in Reynolds number, further reinforces the equivalent length method as a very useful technique for preliminary and non-mission critical calculations.

There is another consideration of the flow regime that arises out of engineering convention, rather than from fundamentals. Strictly, the velocity head (the kinetic energy term in the Bernoulli equation) should be expressed as  $(\alpha v^2/2g)$ . The correction factor,  $\alpha$ , is required because by convention the velocity is taken as the average velocity (i.e.  $v = \text{flow rate} / \text{cross sectional area}$ ). In reality  $(\text{average velocity})^2$  is not equal to  $(\text{average of } v^2)$  and the correction factor is used to avoid having to integrate to get the true average. In turbulent flow  $\alpha$  is very close to 1 and in laminar flow it has a value of 2.

It was stated in section 2 above that to calculate the pressure drop in straight pipe the velocity head is multiplied by the factor  $(fL/D)$ . There is no  $\alpha$  in the Darcy-Weisbach formula (Equation (1)), so what do we do for laminar flow? The answer is that by engineering convention the effect of  $\alpha$  is absorbed into the friction factor. We could include  $\alpha$  and use a friction factor that is only half the usual value, but to keep the arithmetic easy  $\alpha$  is absorbed into the friction factor,  $f$ , and the velocity head is taken as  $(v^2/2g)$ .

A similar thing is done with the resistance coefficients (K values) for pipe fittings. We define the K values to include the value of  $\alpha$  just to keep the arithmetic easy.

There is one exception when it comes to minor losses. What is often called the "exit loss", but which is more accurately the acceleration loss, is the kinetic energy in the stream issuing from the discharge of the pipe. This energy is lost and is equal to one velocity head. There is no way of getting away from it that here you have to use the correct value of  $\alpha$  to get the "exit loss" correct. The only alternative would be to define it to have a K value of 2 in laminar flow, but it would then appear that in laminar flow you lose 2 velocity heads.

In practice this is usually not important. In laminar flow the velocity is low enough that one velocity head is insignificant - and even if doubled with an  $\alpha$  value of 2, it is still insignificant. The K values of fittings in laminar flow can go into the hundreds, or even thousands, and one measly little 2.0 isn't going to bother anybody.

#### 3.4.4 Effect of the fitting roughness

The main causes of the pressure losses in pipe fittings are the changes in direction and cross sectional area. Both of these changes result in acceleration of the fluid and this consumes energy. There will of course be some influence of the friction between the inner surface of the fitting and the fluid on the pressure drop through the fitting, but it needs to be seen in context. Sticking with the example of the L.R. bend, the flow path through the bend can be calculated to be approximately 2.5 times the inside diameter of the pipe.

The equivalent length of a long radius bend is usually taken (perhaps a bit conservatively) as 16. If the overall pressure drop is equivalent to a pipe length of 16 diameters, and the pressure drop due to the actual flow path length (which is affected by the roughness) is equivalent to only 2.5 diameters then it can be seen that a small change in the wall friction inside the bend will have a very small effect on the total pressure drop. In a higher resistance fitting like a globe valve or strainer the effect of the friction is even less.

Experimental work on flow in bends has shown that the roughness does have a measurable impact on the pressure drop. But the experimental work also shows that there are measurable differences in the pressure drop through supposedly identical fittings from

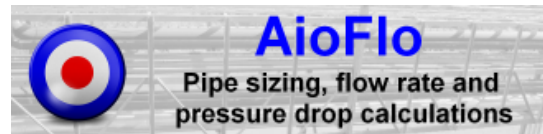
different manufacturers. Because the differences are small, all the generally accepted methods have ignored the roughness in the fitting and have rather selected slightly conservative values for  $(L_e/D)$  and  $(K)$ .

### 3.5 Conversions between the resistance coefficient (K) and the valve flow coefficient ( $C_v$ )

In order to be able to convert between  $K$  and  $C_v$  values it is first necessary to re-arrange Equations (3) and (6) to be in similar units. Equation (3) is in the form of a head of fluid while Equation (6) is in pressure terms. The relation  $\Delta P = \rho gh$  can be used to bring the two equations into equivalent forms. Similarly, the velocity term in Equation (3) can be substituted by volumetric flow/area and the area can of course be expressed in terms of the pipe diameter. Once all these transformations, and a few unit conversions, have been done the relationship becomes:

$$C_v = 29.9 D^2 / \sqrt{K} \quad \text{Equation (10)}$$

where  $D$  is in inches and  $C_v$  is based on US gallons.



## 4. The Crane "2 friction factor" Method for Determining the Resistance Coefficient (K)

There is no doubt that the Crane TP 410 "Flow of Fluids through Valves, Fittings and Pipe" manual has played a major role in the improvement in the quality of hydraulic designs for piping over the last 7 decades. In pointing out some of the weaknesses of the Crane method this section is not aimed at detracting from the enormous contribution made by Crane, but rather to highlight those areas where the state of the art has advanced in the meantime and where engineers involved in pipe flow rate, pipe sizing and pipe pressure drop calculations can take advantage of more accurate methods now available.

Prior to 1976, Crane TP 410 used the equivalent length method for calculating the pressure drops through fittings. The switch to using resistance coefficients ( $K$ ) was made because they believed that the equivalent length method resulted in overstated pressure drops in the laminar flow regime (which is partially true).

Crane found that in fully turbulent flow conditions the resistance coefficient ( $K$ ) for many fittings varied with pipe diameter at exactly the same rate at which the friction factor for clean commercial steel pipe varied with diameter. This is shown in Figure 2-14 of Crane TP 410 (1991). In fully turbulent flow the friction factor  $f_T$  is a function of  $\epsilon/D$  (i.e. roughness/diameter) only, and since  $\epsilon$  is fixed by the assumption of clean commercial steel pipe  $f_T$  becomes a function of pipe size only. Crane never stated that lower values of  $f_T$  in larger pipes were the **cause** of the decrease in the resistance factor  $K$ , but it is common for people to forget that correlation does not imply causation.

It is difficult to understand why, but Crane believed that the resistance factors ( $K$ ) that were determined in this way would be constant **for all flow rates** for a given size of fitting. This was a strange conclusion to come to because data for laminar flow had started appearing from around 1944, and by 1963 it was well enough known and accepted to be mentioned in the 4<sup>th</sup> Edition of Perry's Chemical Engineers' Handbook.

Crane took advantage of the relationship between the equivalent length  $(L_e/D)$  and resistance coefficient ( $K$ ) as shown in Equation (9) above to determine the new  $K$  values from their previously determined and reported equivalent length  $(L_e/D)$  values. The  $(L_e/D)$  values that had been accumulated by Crane had all been measured under conditions of fully turbulent flow, and expressed in terms of length of clean commercial steel pipe. They therefore used  $f_T$ , the Moody friction factor for fully turbulent flow in clean commercial steel pipe of the applicable diameter to convert the equivalent length  $(L_e/D)$  values to resistance coefficient ( $K$ ) values.

The TP 410 manual makes it very clear that the resistance coefficient ( $K$ ) values are to be regarded as constant for all flow rates, and that only the friction factor for fully turbulent flow in clean commercial steel pipe  $f_T$  should be used in the conversion from the old equivalent length  $(L_e/D)$  values. This was because they believed that the equivalent length  $(L_e/D)$  values that they had

determined previously were valid only for fully turbulent flow, but that once they were converted to resistance coefficient (K) values they were applicable to all flows.

Although the link between equivalent length ( $L_e/D$ ) and resistance coefficient (K) was clearly stated to be  $f_T$ , many engineers took it to be just  $f$ , or the friction factor in the connected piping and these engineers used this relationship to generate K values for use in smooth pipe and for lower Reynolds Numbers. Although both of these cases are in contradiction to what Crane intended, one is a valid calculation while the other is wrong. This is the confusion between correlation and causation mentioned earlier.

As was shown above in section 3.4.1, when working with smooth pipe the resistance coefficient (K) for the fitting remains the same but the equivalent length ( $L_e/D$ ) changes. It is therefore wrong to take the Crane ( $L_e/D$ ) values and use the lower friction factor in smooth pipe to generate a lower resistance coefficient (K) from Equation (9). Connecting a fitting to a smooth pipe does not decrease the resistance of the fitting.

On the other hand, it was shown in section 3.4.3 that at lower Reynolds numbers both the friction factor and the fitting resistance coefficient (K) increase, while the equivalent length ( $L_e/D$ ) of the fitting remains constant. It is therefore a valid calculation to take the Crane ( $L_e/D$ ) values and to use the actual friction factor  $f$  at the lower flow rate to generate a new (higher) resistance coefficient (K) value, although this is not how Crane intended their method to be used.

In essence, Crane took Equation (2) and modified it by applying the actual friction factor,  $f$ , in the pipe to the pipe flow (which is obviously the right thing to do) while applying the friction factor for fully turbulent flow in clean commercial steel pipe,  $f_T$ , to the equivalent lengths of the fittings. This is shown in Equation (11):

$$h_L = \left( \frac{fL}{D} + f_T \sum \frac{L_e}{D} \right) \left( \frac{v^2}{2g} \right) \quad \text{Equation (11)}$$

This is why the Crane method is sometimes called the "two friction factor" K method. This also resulted in some engineers developing the misunderstanding that the  $f_T$  friction factor was somehow directly associated with the fitting, and because the fitting had a friction factor it also had a roughness. You will find statements like "*You must not mix the friction factor for a fitting with the friction factor of a pipe*" in the engineering forums on the internet, bearing testament to the belief that fittings somehow have friction factors. Crane never intended people to associate friction factors with fittings, but Crane's intentions have been misunderstood by many.

The result of the switch from the equivalent length ( $L_e/D$ ) method to the resistance coefficient (K) method was (apart from the confusion caused) that while the ( $L_e/D$ ) method may have overstated pressure drops slightly in the laminar flow regime, the new constant K value method horribly understated them. The examples in sections 3.4.2 and 3.4.3 show how the resistance coefficient (K) for a L.R. bend can increase from around 0.2 to 8.2 when the Reynolds Number drops to 100. Fortunately this error is usually not significant in practice because the pressure drops through the fittings tend to be a small part of the overall pressure drop, and a large error in a small portion becomes a small error overall.

When Crane first published their piping design guidelines in 1935, industrial piping was manufactured almost exclusively from carbon steel and the Crane methods were aimed at providing reliable design methods for that pipe. Also, the overwhelming majority of industrial pipe flow is in the turbulent flow regime. Crane certainly succeeded in establishing a comprehensive and accurate design method for turbulent flow in steel pipe. In modern times with the ever increasing use of smooth plastic and high alloy pipe it is essential that engineers fully understand the design methods they use, and that they employ the right method for the problem at hand. The right methods are available in the 2-K and 3-K resistance coefficient methods discussed earlier, and it is time for the piping design world to break with the past and to embrace the new methods.

## 5. Accuracy

Much of what has been said above could be seen to imply that determining the pressure losses in pipe fittings is an exact science. It is not. Very few sources of equivalent length ( $L_e/D$ ) or resistance coefficient (K) values give accuracy or uncertainty limits. A notable exception is the Hydraulic Institute's Engineering Data Book. At the very best the uncertainty would be 10% and in general 25 to 30% is probably a more realistic estimate.

Standard fittings like elbows and tees vary from manufacturer to manufacturer and a tolerance of 25% should be assumed in calculations. Precision engineered items like control valves and metering orifices will of course have much tighter tolerances, and

these will usually be stated as part of the accompanying engineering documentation.

An area that needs particular care is using generic data for proprietary items. Many of the data tables include values for proprietary items like gate, globe, butterfly and check valves, strainers and the like. The actual flow data can vary very widely and variations of -50% to +100% from generic data can be expected.

## 6. Conclusion

At some point in the past the equivalent length ( $L_e/D$ ) method of determining the pressure drop through pipe fittings gained the reputation of being inaccurate. This was quite likely a result of Crane dropping this method in favour of the resistance coefficient (K) method. Recently this attitude has changed in some circles, and hopefully the analysis done above will help convince more design engineers that the equivalent length ( $L_e/D$ ) method is actually very useful and sufficiently accurate in many situations. However, this method does suffer from two serious drawbacks. These are the necessity of defining the pressure drop properties of the fitting in terms of an arbitrary external factor (i.e. the attached piping) and the inability of this method to cope with entrances, exits and fittings with two characteristic diameters (e.g. changes in diameter and orifices). For these reasons the resistance coefficient (K) method is the better route to accurate and comprehensive calculations.

Darby's 3-K method has the capability of taking the fitting size and the flow regime into account. The quantity of data available is gradually increasing and is now roughly equivalent in scope to the Crane TP 410 database. Already some of the higher end software has switched to using Darby's method, and it can be expected that with time it will become more widely used.

The data in Crane TP 410 remains a very valuable resource, but it should be used with an understanding of its range of applicability. Fortunately this data is at its most accurate in the zone of fully turbulent flow, which is where most piping operates. The errors introduced by this method when the flow rate is below the fully turbulent regime can be large relative to the losses in the fittings themselves, but since these are often a small part of the overall losses the errors are often insignificant. As always, an appreciation for the accuracy of the methods being employed enables the engineer to achieve a safe and economical design.

## 7. References

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